

RESEARCH



Real-world effectiveness of Semaglutide treatment on weight loss maintenance after weight loss in patients with obesity or overweight and diabetes

Michela Del Prete¹ · Lidia Gavazzi¹ · Olga Eugenia Disoteco¹ · Federico Vignati¹ · Gianleone Di Sacco¹ · Fabrizio Muratori¹

Received: 18 November 2024 / Accepted: 23 December 2024
© The Author(s) 2024

Abstract

Purpose To report data on the real-world effectiveness and safety of injectable (IS) and oral (OS) therapies in obese or overweight diabetes (T2DM) patients on glycometabolic control, weight loss (WL) and weight maintenance after the use of semaglutide.

Methods 175 subjects with obesity or overweight and T2DM were retrospectively assessed. Of these, 129 (75F, 54 M; mean age 61.2 ± 9.8 years) patients were treated with IS and 46 (24F, 22 M; mean age 65.7 ± 12.8 years) with OS for T2DM and WL. At baseline, mean weight (mW) was 101.8 ± 24.6 kg and 95.2 ± 15.0 kg; mean body mass index (mBMI) was 36.7 ± 8.7 kg/m² and 34.3 ± 5.3 kg/m².

Results After 6 months, in the IS group, 127 patients had a mW and mBMI reduction of -10.4 ± 8.1 kg and -3.9 ± 3.0 kg/m². 46 patients in the OS group had a mW and mBMI reduction of -6.7 ± 5.3 kg and -2.6 ± 2.1 kg/m². After 12 months, 102 patients in the IS group achieved a mW and mBMI reduction of -9.3 ± 7.5 kg and -3.4 ± 2.6 kg/m². 44 patients in the OS group of treatment had a mW and mBMI reduction of -10.7 ± 6.5 kg and -3.9 ± 2.4 kg/m².

After 24 months, 92 patients in the IS group of therapy achieved a mW and mBMI reduction of -15.9 ± 11.4 kg and -5.8 ± 3.7 kg/m².

There was a mean percentage reduction of glycated hemoglobin (HbA1C) of $-1.9 \pm 1.4\%$, $-1.5 \pm 1.9\%$, and $-1.5 \pm 1.7\%$ after 6, 12 and 24 months in the IS group. In the OS group there was a mean reduction of $-1.5 \pm 1.6\%$ after 6 months and of $-0.8 \pm 0.6\%$ after 12 months of therapy.

Conclusions Semaglutide induces WL maintenance after 12 and 24 months of treatment. Our results show a comparable effectiveness of IS and OS in T2DM patients with obesity or overweight. IS and OS treatment provide significant WL that allows reaching the glycometabolic therapeutic goal in a short time.

Level of Evidence Level II

Keywords Semaglutide · Overweight · Obesity · Continuing weight loss · Diabetes

Introduction

It is known that throughout the world that approximately 40% of adults are overweight and more than 15% of subjects are living with obesity. Obesity is a complex condition linked to chronic inflammation leading to the development

of comorbidities such as metabolic syndrome, cardiovascular disease, type 2 diabetes (T2DM), non-alcoholic steatohepatitis or different types of tumors [1].

In the last years there has been a significant increase in knowledge of the use of glucagon-like peptide 1 receptor agonist (GLP1-RA) agonists in the treatment of T2DM and obesity or overweight. Since 2005, several GLP1-RA agonists have been approved for the treatment of T2DM, but only liraglutide and semaglutide have been approved for the treatment of obesity. Semaglutide is a long-acting GLP1-RA, indicated for the management of adults with T2DM and overweight or obesity as an adjunct to diet and exercise as

✉ Michela Del Prete
michela.delprete@asst-lariana.it

¹ Division of Endocrinology, Diabetology and Clinical Nutrition, Sant'Anna Hospital – ASST Lariana, Como, Italy

shown by the SUSTAIN trial program [2]. GLP1-RA have demonstrated effectiveness in lowering glycated hemoglobin (HbA1C), decreasing weight, and reducing the risk of major cardiovascular events in patients with T2DM.

Since 2021, Semaglutide 2.4 mg has been approved by the European Medicines Agency (EMA) for the long-term treatment of both non-diabetic adult and adolescent subjects with obesity or overweight [3]. Currently, Semaglutide is the only GLP-1-RA available in both injectable and oral form for the treatment of T2DM subjects [4]. The STEP 2 study has shown that the once weekly Semaglutide therapy at a dose of 2.4 mg in T2DM subjects with overweight or obesity was more effective in weight loss (WL) compared to Semaglutide at dose of 1.0 mg or placebo [5]. The effectiveness of Semaglutide treatment at a dose of 2.4 mg on WL maintenance in subjects with overweight or obesity was also reported in the STEP 4 and STEP 5 studies respectively over the 48 and 104 weeks of therapy [6, 7]. Oral Semaglutide 50 mg also induced a significant WL compared to placebo (15.1% versus 2.4%) [8], however the use in clinical practice of oral Semaglutide 50 mg is not yet approved.

Here, we report the efficacy and safety of semaglutide therapy on glycemic control, the effectiveness on WL maintenance of Semaglutide therapy, and the comparison of efficacy between injectable (IS) and oral Semaglutide (OS) treatment alone or in combination with other antidiabetic drugs in T2DM subjects with obesity or overweight. Preliminary data from this article were reported at the 31st European Congress on Obesity (ECO 2024) [9, 10].

Patients and methods

One hundred and seventy-five subjects with obesity or overweight and recent diagnosis of (<6 months) T2DM were retrospectively evaluated for WL and glycemic control with Semaglutide therapy at the Sant'Anna Hospital of Como. Baseline patient characteristics are reported in Table 1. At baseline, 35 patients had arterial hypertension in the IS group and 11 in the OS group. At the time of the first visit, no patient had other known comorbidities.

All patients were aged more than 18 years old and examined every 6 months from the beginning of the treatment. Of these, 129 subjects (75 Female and 54 Male; mean age 61.2 ± 9.8 years) were treated with IS and 46 (24 Female and 22 Male; mean age 65.7 ± 12.8 years) with OS therapy for WL and glycemic control.

Semaglutide therapy was administered as monotherapy or in combination with metformin. The patients were not taking other antidiabetic drugs. IS treatment was administered once weekly subcutaneously. The starting IS dose was of 0.25 mg and with monthly increases up to 1.0 mg. OS therapy was administered once daily at the starting dose of 3 mg and with

Table 1 Baseline characteristics of patients treated with injectable and oral Semaglutide

Overall patients (175)	IS (n=129)	OS (n=46)
Age (years; mean $\pm$ SD)	61.2 ± 9.8	65.7 ± 12.8
Sex (n)		
F	75	24
M	54	22
Weight (Kg; mean $\pm$ SD)	101.8 ± 24.6	95.2 ± 15.0
BMI (kg/m^2 ; mean $\pm$ SD)	36.7 ± 8.7	34.3 ± 5.3
HbA1C (%; mean $\pm$ SD)	7.7 ± 1.6	7.6 ± 1.6
Concomitant anti-diabetic medications (n)		
Metformin	97	39
Known comorbidities at first visit (n)		
Arterial hypertension	35	11

IS injectable Semaglutide, OS oral Semaglutide, BMI body mass index, HbA1C glycated hemoglobin, SD standard deviation

monthly increases up to 14 mg in all patients. Data on semaglutide effectiveness and safety was recorded up to 24 months for IS treatment group and up to 12 months for OS treatment group from baseline.

The percentage of WL and the mean body mass index m(BMI) changes in all patients and subgroups according to the degree of obesity were calculated to assess the effectiveness of both IS and OS group therapies. Percentage changes in glycated hemoglobin (HbA1C) were also calculated from baseline in both subject groups of treatment.

All patients were required to adhere to dietary and behavioral advice (as 200 min of walking per week) and any concomitant drug treatment. Patients received the total daily energy requirement calculated by using Harris-Benedict formula. The aim was to receive the target energy per day as a 500-kcal subtraction from the total energy requirement. The diet regime included approximately 50% carbohydrates, 30% lipids and 20% proteins. Patient consents were not required due to the retrospective nature of this study.

Statistical analysis was performed with GraphPad Prism software (version 9, GraphPad Software Inc., La Jolla, CA) and as also reported elsewhere [11]. Data were expressed as mean $\pm$ standard deviation (SD) and p-value was considered significant if <0.05 . The Kolmogorov–Smirnov test was used to evaluate data distribution. The comparison of data on weight and BMI changes was performed using the two-tailed paired student's t or Wilcoxon test, Pearson correlation or Spearman tests.

Results

Effectiveness assessment of weight loss and weight loss maintenance after IS and/or OS

At the first visit, mean weight m(W) was 101.8 ± 24.6 and kg 95.2 ± 15.0 kg; mBMI was 36.7 ± 8.7 kg/m² and 34.3 ± 5.3 kg/m² respectively for IS and OS group of subjects. At baseline there were no statistical differences mW ($p=0.089$) and mBMI ($p=0.39$) between the two groups.

After 6 months from the beginning of therapy, 127/129 patients treated with IS (2 patients have not yet reached 6 months of treatment) achieved a mW of 95.4 ± 24.1 kg and mBMI of 34.2 ± 9.4 kg/m², with a mW and mBMI reduction respectively of -10.4 ± 8.1 kg ($-9.5 \pm 6.5\%$) and -3.9 ± 3.0 kg/m². 46/46 patients treated with OS had a mW of 90.3 ± 12.7 kg and a mBMI of 33.4 ± 4.3 kg/m². The mW reduction was -6.7 ± 5.3 kg ($-6.8 \pm 5.8\%$); while the mBMI reduction was -2.6 ± 2.1 kg/m².

After 12 months of therapy, 102/129 subjects achieved a mW of 88.5 ± 17.7 kg and mBMI of 31.9 ± 5.5 kg/m², with a mW and mBMI reduction respectively of -9.3 ± 7.5 kg ($-9.2 \pm 7.7\%$) and -3.4 ± 2.6 kg/m². 44/46 patients treated with OS achieved a mW of 89.2 ± 14.0 kg and mBMI of 31.5 ± 4.3 kg/m² with a mWL of -10.7 ± 6.5 kg ($-10.4 \pm 5.8\%$); and mBMI reduction of -3.9 ± 2.4 kg/m² (Figs. 1 and 2).

There was no statistical difference in mWL variations and mBMI changes after 6 (weight $p=0.092$; BMI $p=0.092$) and 12 months (weight $p=0.73$; BMI $p=0.74$) of therapy between IS and OS therapy.

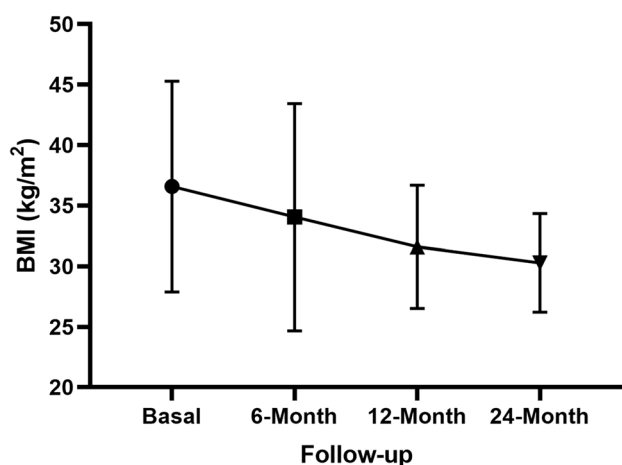


Fig. 1 Continuing weight loss of patients treated with Injectable Semaglutide. BMI body mass index. Statistical significance from baseline: 6 months $p<0.0001$, 12 months $p<0.0001$ and months $p<0.0001$

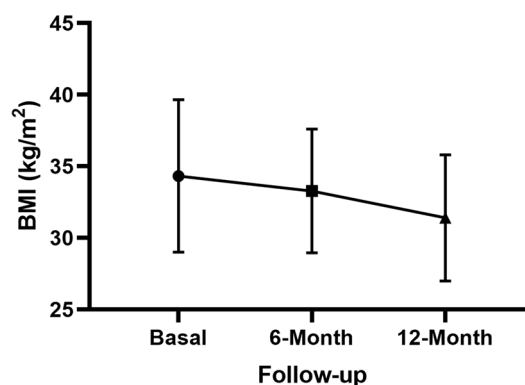


Fig. 2 Weight loss of patients treated with Oral Semaglutide. BMI body mass index. Statistical significance from baseline: 6 months $p<0.0001$ and 12 months $p<0.0001$

After 24 months from the beginning of the therapy, only patients of the IS group were assessed. Of these, 92/129 subjects achieved a mW of 83.2 ± 14.0 kg and mBMI of 30.6 ± 4.4 kg/m², with a mW and mBMI reduction respectively of -15.9 ± 11.4 kg ($-15.0 \pm 8.7\%$) and -5.8 ± 3.7 kg/m² (Fig. 1).

In the IS group there was a statistical significance in mW and mBMI reduction after 6 (W $p<0.0001$; BMI $p<0.0001$), 12 (W $p<0.0001$; BMI $p<0.0001$) and 24 (W $p<0.0001$; BMI $p<0.0001$) months of therapy from baseline using the paired Student's t test.

IS and OS therapy were well tolerated, and no patients in both groups experienced severe adverse events like nausea or vomiting. Furthermore, no patient had optic neuropathy.

Effectiveness on glycemic control

By considering the glycemic control, at the first visit mean HBA1C was of $7.7 \pm 1.6\%$ and after 6, 12, and 24 months from the beginning of the treatment was respectively $6.4 \pm 0.8\%$, $6.3 \pm 0.6\%$, and $6.4 \pm 1.0\%$, with a mean percentage reduction of $-1.9 \pm 1.4\%$, $-1.5 \pm 1.9\%$, and $-1.5 \pm 1.7\%$ in the IS group (Fig. 3). There was a statistical significance in mean HBA1C reduction after 6 ($p<0.0001$), 12 ($p<0.0001$) and 24 ($p=0.0024$) months of therapy from baseline.

In the OS group at baseline mean HBA1C was $7.6 \pm 1.6\%$. After 6 months of therapy HBA1C was $6.4 \pm 0.6\%$, with a mean reduction of $-1.5 \pm 1.6\%$. After 12 months of OS treatment HBA1C was $6.3 \pm 0.7\%$, with a mean reduction of $-0.8 \pm 0.6\%$ (Fig. 4). There was a statistical significance in mean HBA1C reduction after 6 ($p=0.0010$) and 12 ($p=0.01$) months from baseline.

By comparing the mean percentage reduction of HBA1C of IS and OS group, there were no statistical differences

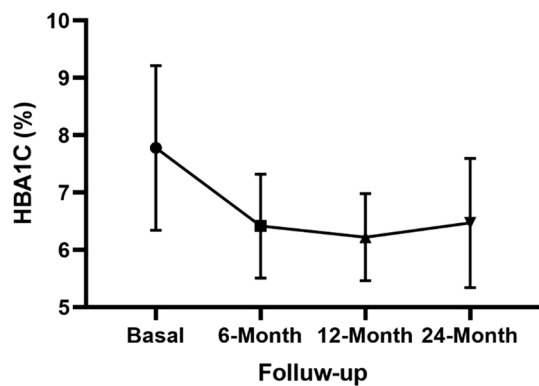


Fig. 3 Efficacy of Injectable Semaglutide on glycemic control. *HbA1C* glycated hemoglobin. Statistical significance from baseline: 6 months $p < 0.0001$, 12 months $p < 0.0001$ and 24 months $p = 0.0024$

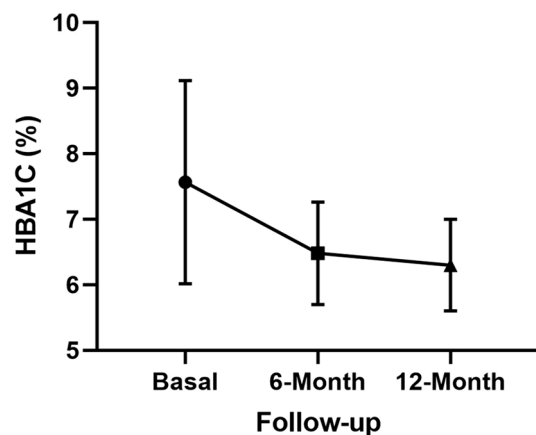


Fig. 4 Efficacy of Oral Semaglutide on glycemic control. *HbA1C* glycated hemoglobin. Statistical significance from baseline: 6 months $p = 0.0010$ and 12 months $p = 0.01$

at baseline and after 6 and 12 months of therapy (baseline $p = 0.088$; 6 months $p = 0.089$; 12 months $p = 0.18$).

Discussion

The use of semaglutide for the treatment of T2DM subjects was based on the results of the clinical development program called SUSTAIN, which enrolled more than 8000 diabetic subjects. All these studies showed a weight loss markedly superior to the comparison group [2].

A multicenter, phase 2 study on subject with obesity was then performed: 957 people were randomized to the treatment with doses of semaglutide between 0.05 mg, 0.1 mg, 0.2 mg, 0.3 mg and 0.4 mg/day or placebo. Liraglutide 3.0 mg/day was included for comparison. Weight loss during the treatment with semaglutide at week 52

was 16.2% compared to baseline at the dose of 0.4 mg/day, versus -2.3% for placebo. To provide a comparison on efficacy, patients on liraglutide 3.0 mg, at week 52, had similar weight loss as those on 0.2 mg of semaglutide [12]. The clinical effectiveness of semaglutide treatment in the management of obesity was then confirmed by the results of Phase 3 Trials: Semaglutide Treatment Effect in People with Obesity (STEP program) [3, 5, 6, 13].

In this study we verified in real-world the efficacy of semaglutide both subcutaneously and orally in T2DM subjects with obesity or overweight. We treated diabetic patients with a personalized nutritional intervention, advice and information material for physical exercise according to the characteristics and possibilities of each patient.

This program has certainly increased the result on weight loss and improvements in glyco-metabolic parameters. One of the reasons that has most disappointed doctors in the management of obesity is the weight regaining after weight loss: this aspect is one of the most controversial in the long-term management of the obese patient.

Our results demonstrated comparable effectiveness both with IS and OS treatment in T2DM adults with obesity and overweight. In our study both therapies provide significant loss of weight allowing patients to reach the glycol-metabolic therapeutic goal in a short time.

In the treatment of overweight T2DM patients, therapy should continue throughout life (at least in patients without disease remission) and long-term studies in this population can help us understand whether maintaining weight after weight loss is an achievable goal.

Recently, in a fundamental study on obese population with pre-existing cardiovascular diseases (Select study) semaglutide determined a weight loss up to approximately the 65th week and this result on weight loss was maintained after 4 years from the beginning of the therapy with semaglutide [14]. For the first time, therefore, weight maintenance is more easily manageable in both overweight and obese T2DM patients. Further studies will be necessary to understand what the dosage will be to maintain weight. In obese patients, further studies should probably be conducted to verify whether the weight maintenance dosage was related to the starting BMI. Other studies will also be necessary to establish whether optimal therapy should be chronic.

Limitations of this study include its retrospective nature as well as the different samples of patients enrolled in the two analyzed subgroups. This difference is also due to the different entry into the market in our country of oral semaglutide compared to the injectable formulation. This allowed us to report data only at 12 months for oral semaglutide from baseline. Furthermore, no patients were excluded from the study according to the retrospective nature of the study, however not all patients treated with IS or OS have reached

6-, 12- or 24-month follow-up considered for the statistical analysis.

In conclusion, in our study, semaglutide treatment confirmed what was reported above. In fact, our results demonstrated the effectiveness and safety over 24 months of semaglutide treatment in T2DM obese subjects on loss of weight and HBA1C improvement. In line with the results of the STEP 4 study, semaglutide therapy induces weight loss and continuing weight loss maintenance after 12 and 24 months of treatment in subjects with obesity or overweight.

Author contributions All authors contributed to the study concept and design. Material preparation, data collection and analysis were performed by Michela Del Prete, Fabrizio Muratori and Lidia Gavazzi. The first draft of the manuscript was written by Fabrizio Muratori and Michela Del Prete. All the authors commented on previous versions of the manuscript. All authors read and approved of the final manuscript.

Funding The authors declare that no funds, grants, or other support were received during the preparation of this manuscript.

Data availability No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate This is a retrospective observational study, and no ethical approval is required. Informed consent was obtained from all the participants of the study.

Competing interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

References

1. Klook S, Ziegler CG, Dischinger U (2023) Obesity and its comorbidities, current treatment options and future perspectives: challenging bariatric surgery? *Pharmacol Ther* 251:108549
2. Aroda VR, Ahmann A, Cariou B, Chow F, Davies MJ, Jodar E et al (2019) Comparative efficacy, safety, and cardiovascular outcomes with once-weekly subcutaneous semaglutide in the treatment of type 2 diabetes: insights from the SUSTAIN 1–7 trials. *Diabetes Metab* 45(5):409–418
3. Wilding JPH, Batterham RL, Calanna S, Davies M, Van Gaal LF, Lingvay I et al (2021) Once-weekly semaglutide in adults with overweight or obesity. *N Engl J Med* 384(11):989–1002
4. Thethi TK, Pratley R, Meier JJ (2020) Efficacy, safety and cardiovascular outcomes of once-daily oral semaglutide in patients with type 2 diabetes: the PIONEER programme. *Diabetes Obes Metab* 22(8):1263–1277
5. Davies M, Faerch L, Jeppesen OK, Pakseresht A, Pedersen SD, Perreault L et al (2021) Semaglutide 2.4 mg once a week in adults with overweight or obesity, and type 2 diabetes (STEP 2): a randomised, double-blind, double-dummy, placebo-controlled, phase 3 trial. *Lancet* 397(10278):971–984
6. Rubino D, Abrahamsson N, Davies M, Hesse D, Greenway FL, Jensen C et al (2021) Effect of continued weekly subcutaneous semaglutide vs placebo on weight loss maintenance in adults with overweight or obesity: the STEP 4 randomized clinical trial. *JAMA* 325(14):1414–1425
7. Garvey WT, Batterham RL, Bhatta M, Buscemi S, Christensen LN, Frias JP et al (2022) Two-year effects of semaglutide in adults with overweight or obesity: the STEP 5 trial. *Nat Med* 28(10):2083–2091
8. Knop FK, Aroda VR, do Vale RD, Holst-Hansen T, Laursen PN, Rosenstock J et al (2023) Oral semaglutide 50 mg taken once per day in adults with overweight or obesity (OASIS 1): a randomised, double-blind, placebo-controlled, phase 3 trial. *Lancet* 402(10403):705–719
9. Del Prete M DO, Di Sacco G, Vignati F, Gavazzi L, Muratori F. Abstract GC4.192 Comparison of efficacy of injectable and oral Semaglutide treatment in patients with obesity and overweight and with type 2 diabetes. 2024.
10. Del Prete M DSG, Vignati F, Gavazzi L, Disoto O, Muratori F. Abstract GC4.190: Efficacy of Semaglutide therapy on continuing weight loss maintenance after weight loss in patients with diabetes and with obesity. 2024.
11. Del Prete M, Vignati F, Di Sacco G, Gavazzi L, Dellepiane D, Muratori F (2023) Real-life efficacy of liraglutide therapy on weight-loss in patients with overweight and obesity. *J Surg Res* 6:245–250
12. O'Neil PM, Birkenfeld AL, McGowan B, Mosenzon O, Pedersen SD, Wharton S et al (2018) Efficacy and safety of semaglutide compared with liraglutide and placebo for weight loss in patients with obesity: a randomised, double-blind, placebo and active controlled, dose-ranging, phase 2 trial. *Lancet* 392(10148):637–649
13. Wadden TA, Bailey TS, Billings LK, Davies M, Frias JP, Koroleva A et al (2021) Effect of subcutaneous semaglutide vs placebo as an adjunct to intensive behavioral therapy on body weight in adults with overweight or obesity: the STEP 3 randomized clinical trial. *JAMA* 325(14):1403–1413
14. Lingvay I, Brown-Frandsen K, Colhoun HM, Deanfield J, Emerson SS, Esbjerg S et al (2023) Semaglutide for cardiovascular event reduction in people with overweight or obesity: SELECT study baseline characteristics. *Obesity (Silver Spring)* 31(1):111–122

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.